



**INDIAN INSTITUTE OF INFORMATION TECHNOLOGY
SONEPAT**

Submitted By:

Shubham Singh

CSE-B

2nd Semester

12411063

Submitted To:

Vinay Kaushik Sir



Google Summer of Code

sugarlabs



Sugar Labs Website Redesign – Documentation

List of All Important Links:

(ctrl + click to open)

LINK TO THE : [OFFICIAL HOSTED PAGE LINK](#)

- 1) [Github Repository \(Lead by me\).](#)
- 2) [Github Designs Repository \(all designs made by me\)](#)
- 3) [Official Launch Video \(from the point where I started to speak\).](#) (MUST WATCH) :)
- 4) [Launch Promo Video \(made by me\)](#)
- 5) [LinkedIN Post1.](#)
- 6) [LinkedIN Post2.](#)
- 7) [Official GSoC Mail.](#)
- 8) [Official GSoC Project Proposal \(MUST SEE\)](#)

So it all started with me trying to contribute in SugarLabs, Apache Foundation, under open sourced. I was able to market myself well with the contributors and maintainers at SugarLabs as a UI/UX designer, and hence after making 6 contributions in SugarLabs' MusicBlocks (the project in which I've got GSoC internship inside), I got the lead for designing the coding out the ENTIRE SugarLabs website, which originally wasn't well designed.

Number of PRs Merged in Jan – 5

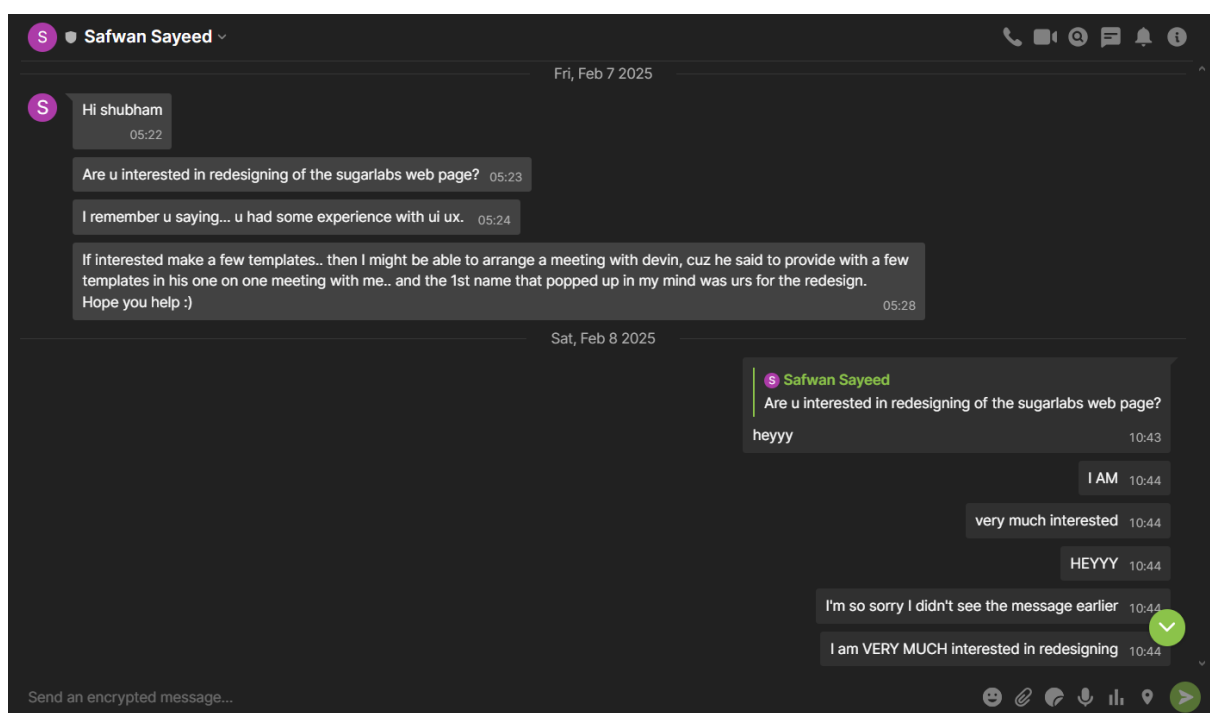
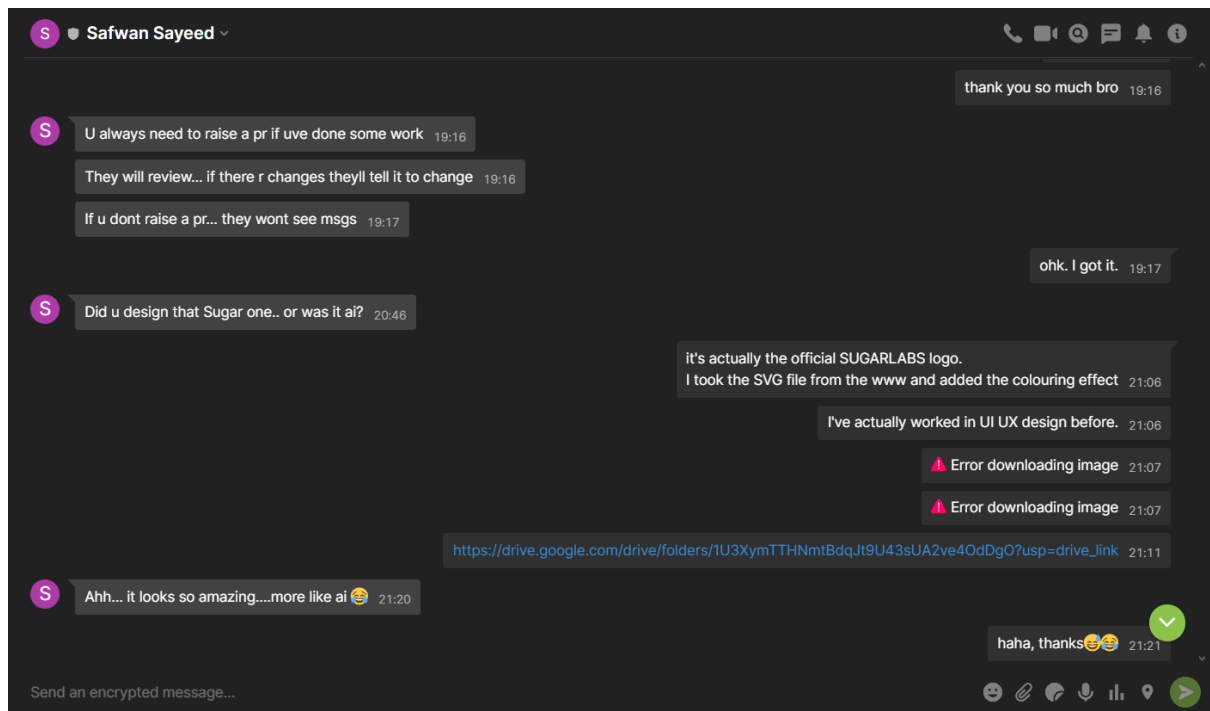
Number of PRs Merged in Feb – 8

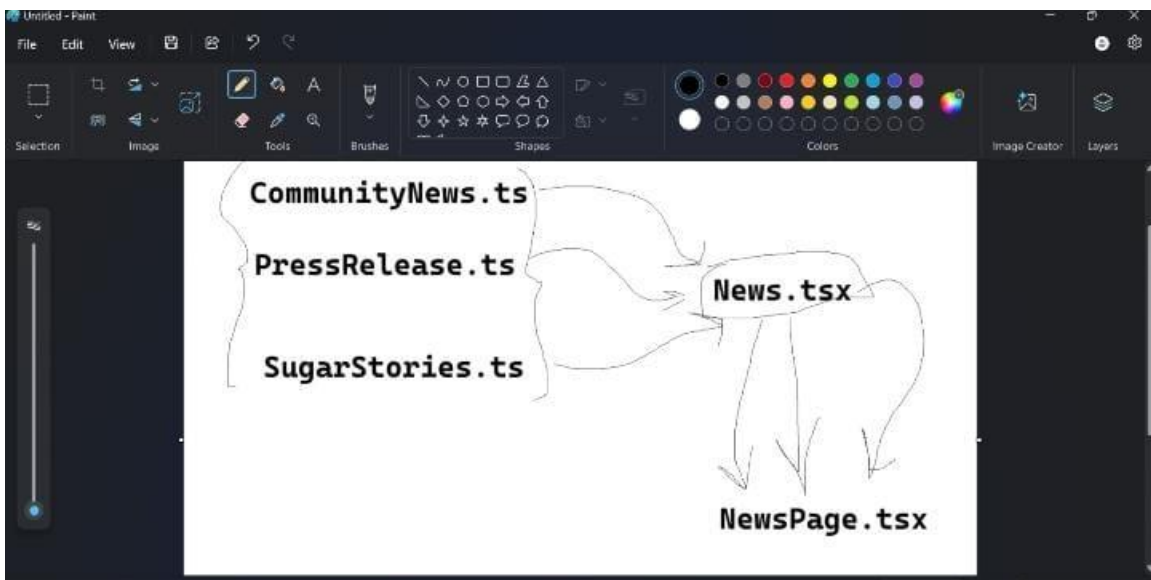
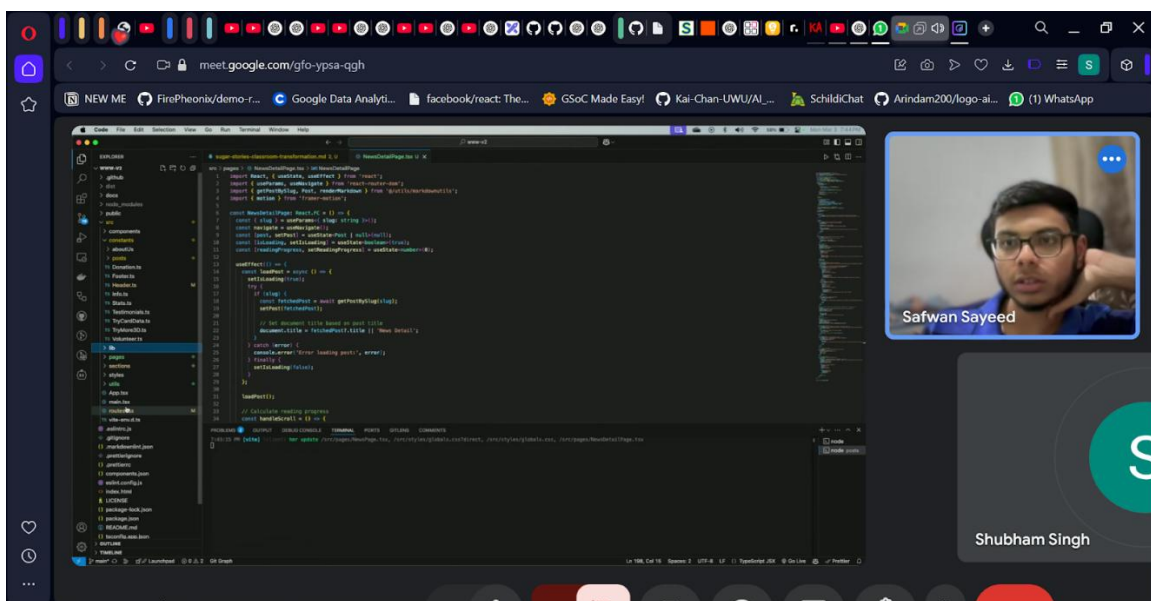
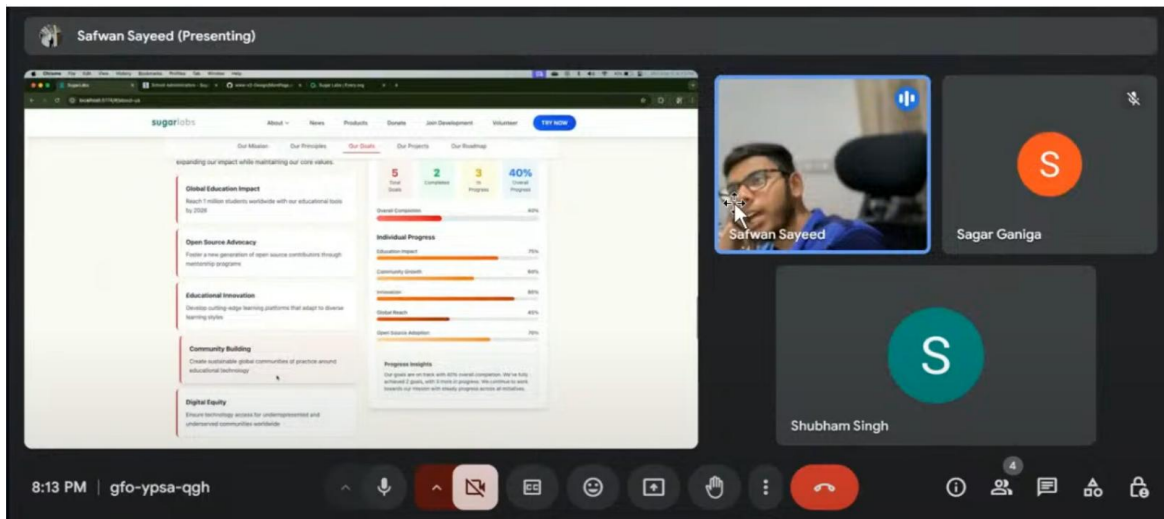
Number of PRs merged in March – 9

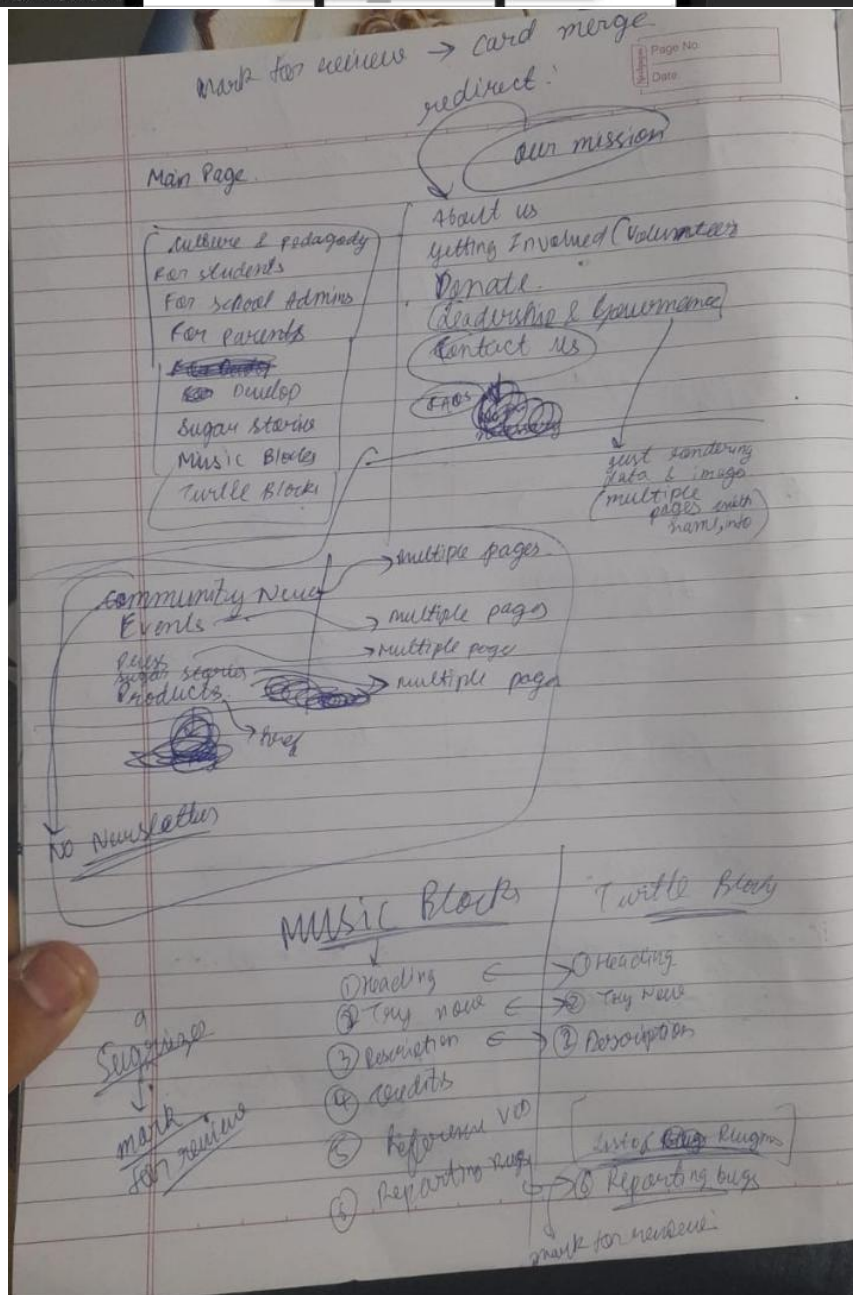
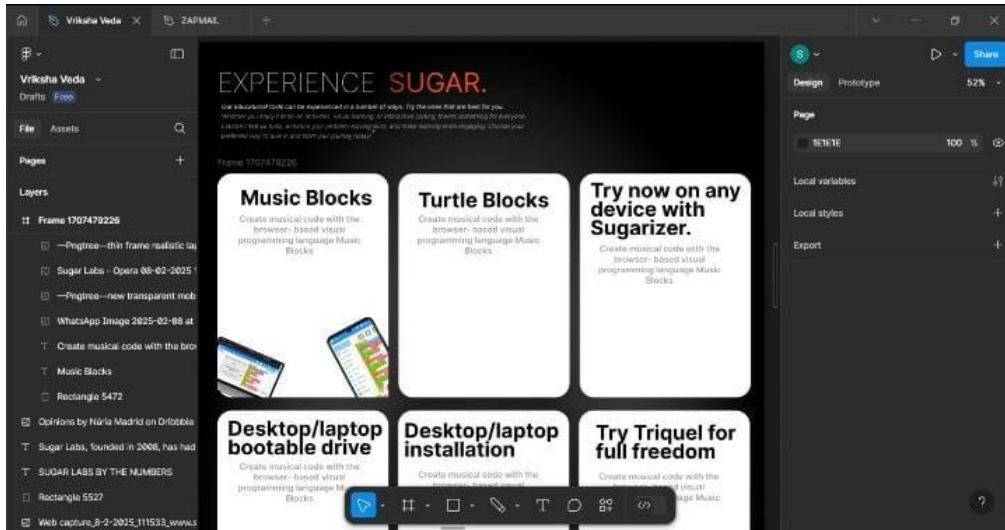
Number of PRs merged in April – 3

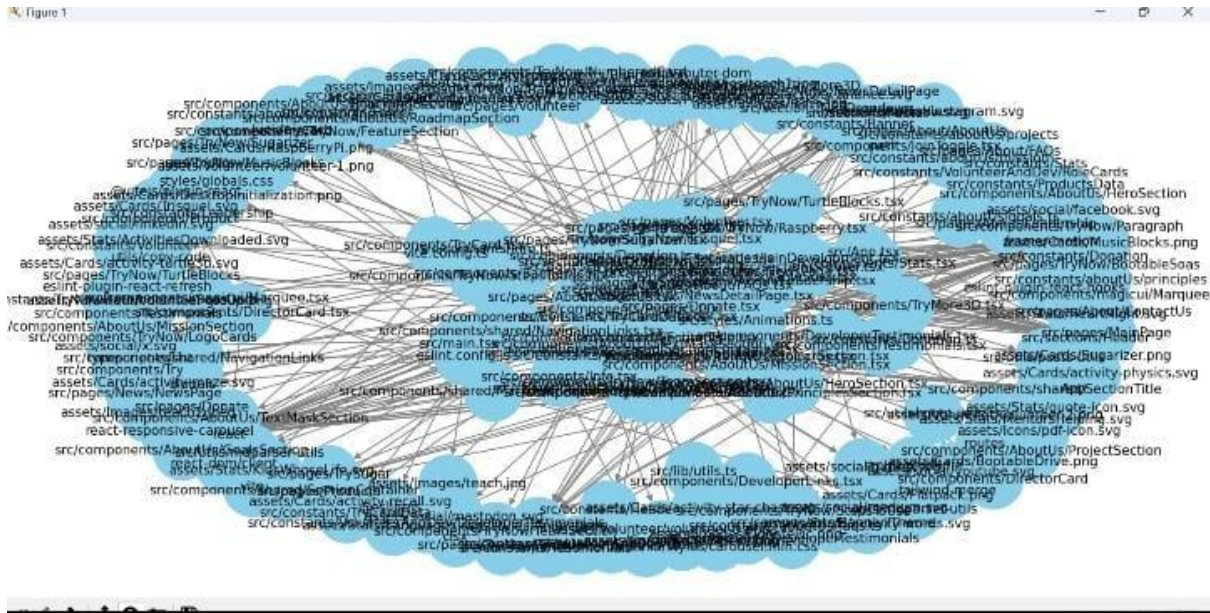
It was relentless journey, with multiple google meets, design discussions, IMAGINE coding out the entire website with over 40,000 lines of code in 3 months along with designing. It was extremely hectic as a first who was also trying to learn how to code along side. But with countless sleepless nights, BARELY 75% attendance in all subjects, and determination to get interned in my first year and shock the world, I MANAGED TO PULL OFF EVERYTHING.

A screenshot of how it all started:

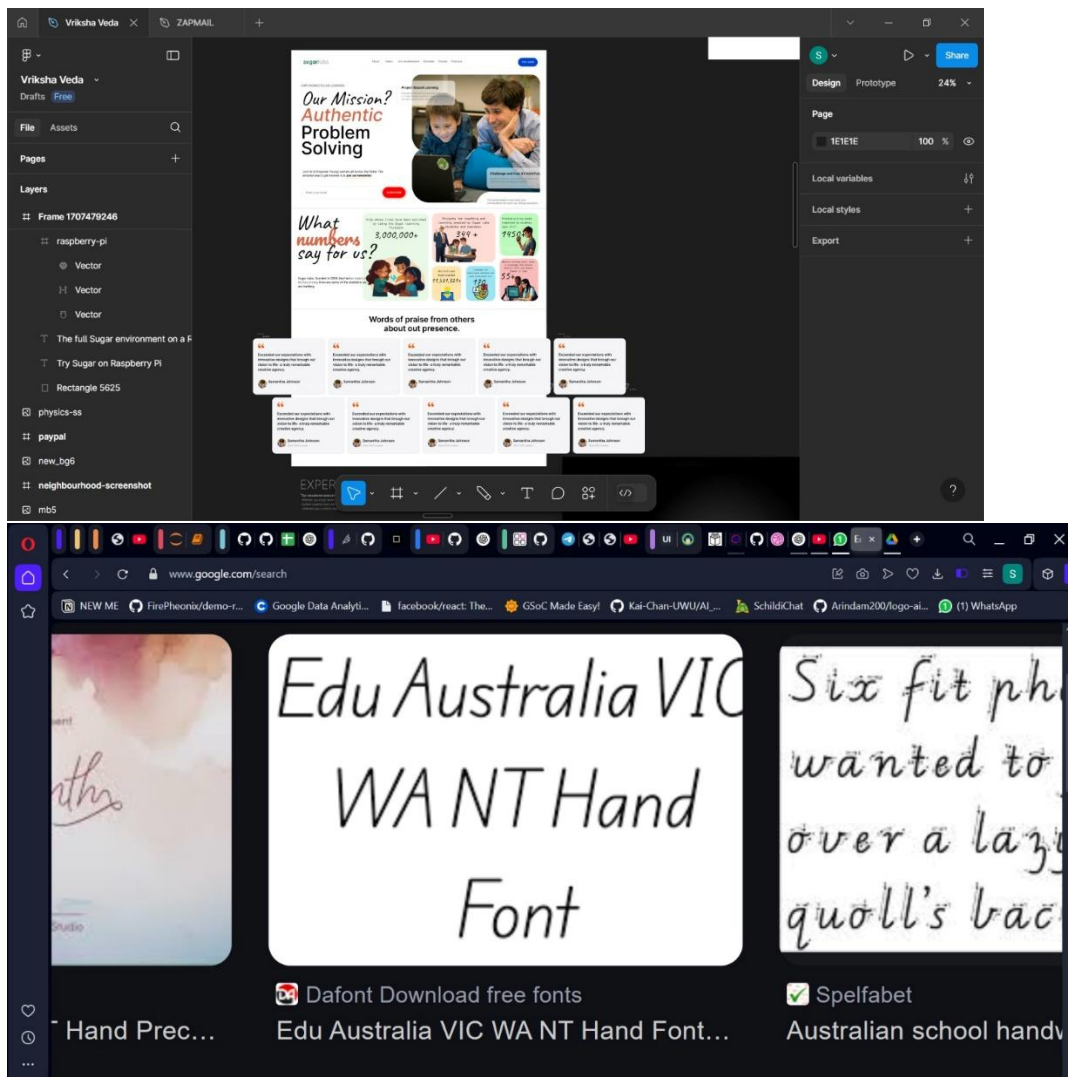






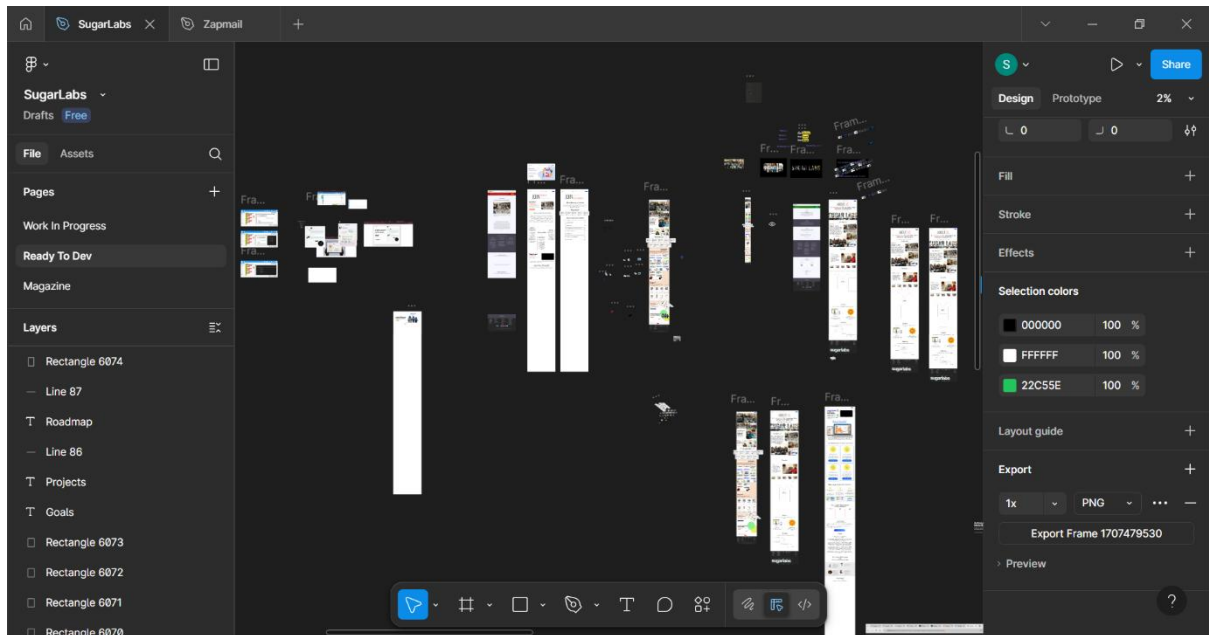


(the above is the MESSED up python script that I ran :))



Role and Contribution

I served as the **Lead Designer** and **Sub-Technical Lead** for the complete redesign and development of the [Sugar Labs official website](#). This is a real-world, open-source project under the Sugar Labs organization, where I worked on both **frontend development** and **UI/UX design**.



This project consists of over **10,000+ lines of code from my side**, and is part of a larger community initiative to modernize the web presence of Sugar Labs — an educational nonprofit that supports learning platforms for children.



I have also been selected as a **Google Summer of Code (GSoC)** contributor with Sugar Labs, making this contribution highly impactful and technically sound.

May 8, 2025

To Whom It May Concern:

Google is pleased to let you know that Shubham Singh has been accepted to participate in the Google Summer of Code 2025 program, which is administered from our Corporate Headquarters at 1600 Amphitheatre Parkway, Mountain View, CA 94043.

Shubham will participate as a GSoC Contributor beginning June 2, 2025 and ending on September 1, 2025, developing open source software for Sugar Labs. Shubham is expected to spend approximately 175 hours working on their project. Shubham will work from home on their project and will not be required to visit either Google or the offices of Sugar Labs.

More information on Shubham Singh's project can be found at: <https://summerofcode.withgoogle.com/programs/2025/projects/N4KQIX8T>. The program is administered by Google, Shubham Singh will not be directly employed by Google during participation in Google Summer of Code.

This letter serves as official proof and verification of Shubham Singh's acceptance into Google Summer of Code 2025. Alternate proof cannot be provided. However, Shubham may, subject to their mentor's approval, make adjustments to the start and end dates of the program, and their mentor may offer to provide verification to supplement this letter.

To learn more about the GSoC program visit our website <https://g.co/gsoc>. Should you require any further information, please do not hesitate to contact our team via email at gsoc-support@google.com

Best,



Stephanie Taylor

Highlights of the Work Done

1. Complete Redesign of UI

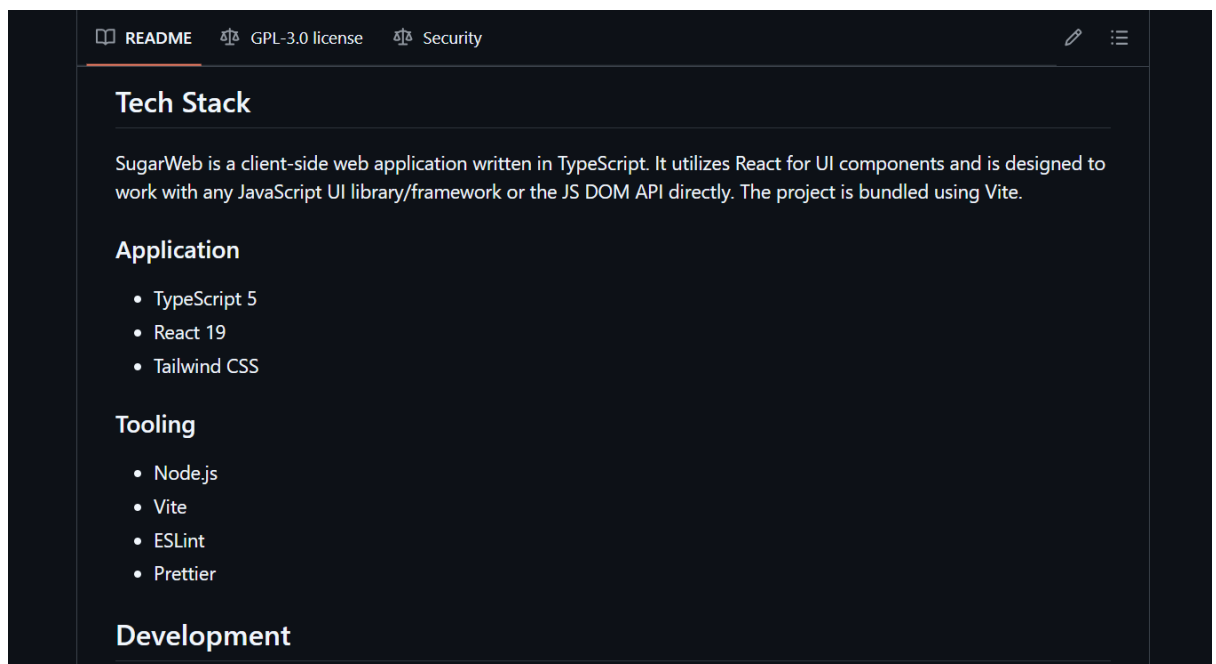
- Reimagined the layout and flow to make it user-friendly and responsive.

- Created modern, accessible, and minimalist designs using **Figma**, which we later coded.
- Focused on contrast, readability, and mobile responsiveness (fully responsive design using TailwindCSS).



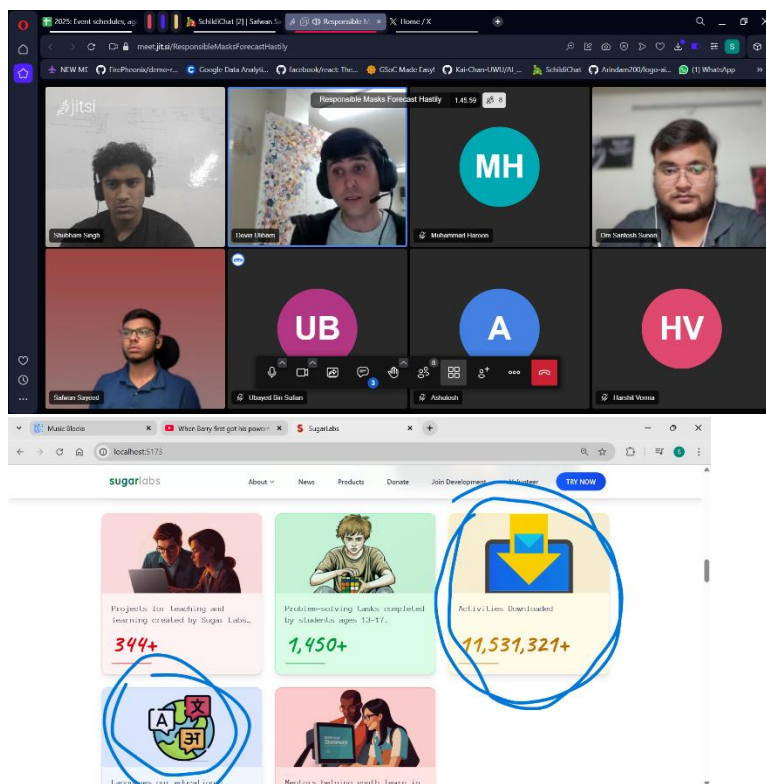
2. Development & Optimization

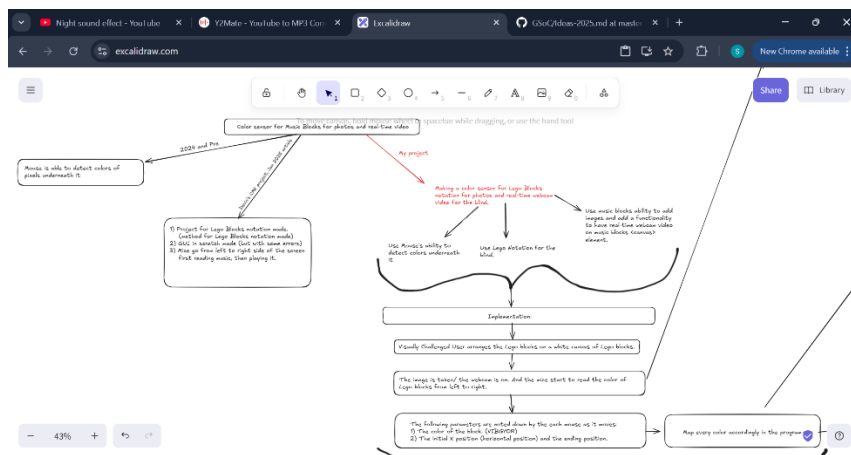
- **Tech Stack Used:** ReactJs, NodeJS, Typescript, ShadCN UIs, TailwindCSS, Markdown, MDX, GitHub CI/CD workflows.
- Implemented lazy loading, image optimization, and smooth routing.



3. Page Contributions

- Created and enhanced the following pages:
 - Homepage
 - Community page
 - Contribute page
 - Project showcase section
 - About us page
 - Contact Us page
 - FAQs page
 - News Page
 - Blogs Page
 - Leadership Page
 - Products Page
- Also added Markdown support to dynamically generate blog/news content.





Music Blocks A Musical Microworld

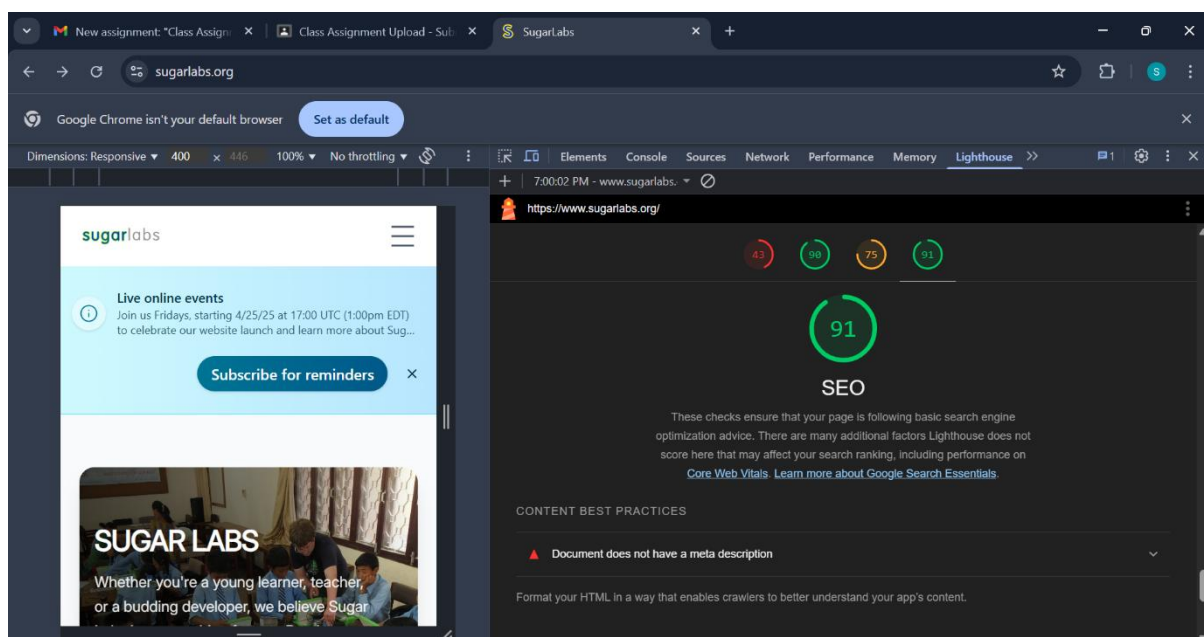
"All musicians are subconsciously
mathematicians" – Monk

Music Blocks is a collection of manipulative tools
for exploring fundamental musical concepts in
an integrative and fun way.



4. Accessibility & SEO Enhancements

- ARIA attributes were added for screen reader support.
- Enhanced semantic HTML structure for better SEO.
- Lighthouse scores improved significantly (SEO: 91%).



5. Version Control & Collaboration

- Used GitHub Projects for task tracking.
- Led and reviewed multiple pull requests (PRs), ensured clean code practices.
- Managed issues and collaborated with other developers across time zones.

< -- AN EXTRACT FROM MY PROJECT PROPOSAL FOR GSOC-- >

I have been an active contributor to Sugar Labs for the past three months. My contributions include:

Number of Total PRs merged: 25

- **Merged 5 PRs in the Music Blocks repository**, included bug fixes, code changes, implementation issues.
- **Merged 20+ PRs in the www-v2 repository.**
- Built entire pages in **www-v2**, improved UI/UX, and worked as the **Design Lead for the same.**
- Contributed to **www2-design**, a repository containing all my complete design assets.
The www-v2 has been completed.

Official Member of SugarLabs organizations

Below are Pull Requests (PRs) created by me: (PRs)

Link to to Issue	Resolution description in short	Link to pull request
	<u>Adding Responsiveness in Blocks issue: #4350</u>	<u>https://github.com/sugarlabs/musicblocks/pull/4350</u>
	<u>Enhancement: utils.js Translation method #4265</u>	<u>https://github.com/sugarlabs/musicblocks/pull/4265</u>

		icblocks/pull/4265
	<u>Fix: Lilypond FileName issue: #4287</u>	https://github.com/sugarlabs/mus-icblocks/pull/4287
	<u>Added link to SugarLabs issue: #4282</u>	https://github.com/sugarlabs/mus-icblocks/pull/4282
	<u>Data Transfer from www1 to www2 issue: #29</u>	https://github.com/sugarlabs/www-v2/pull/29
	<u>Optimized Testimonials Done issue: #14</u>	https://github.com/sugarlabs/www-v2/pull/14
	<u>Adding FAQs section issue: #24</u>	https://github.com/sugarlabs/www-v2/pull/24
	<u>Volunteer and Development Page issue: #32</u>	https://github.com/sugarlabs/www-v2/pull/32
	<u>productsPageDone #37</u>	https://github.com/sugarlabs/www-v2/pull/37
	<u>All Try Now Pages Done #38</u>	https://github.com/sugarlabs/www-v2/pull/38

	<u>DonationSector #59</u>	<u>https://github.com/sugarlabs/www-v2/pull/59</u>
	<u>Floating SVGs added #60</u>	<u>https://github.com/sugarlabs/www-v2/pull/60</u>
	<u>Z-Indexed and percentage widthing #72</u>	<u>https://github.com/sugarlabs/www-v2/pull/72</u>
	<u>SVGs and Navigation #78</u>	<u>https://github.com/sugarlabs/www-v2/pull/78</u>
	<u>MD Files Added and Edited #99</u>	<u>https://github.com/sugarlabs/www-v2/pull/99</u>
	<u>Changed Image #102</u>	<u>https://github.com/sugarlabs/www-v2/pull/102</u>
	Adding Data for Annual Report	<u>https://github.com/sugarlabs/www-v2/pull/104</u>
	widthing of flathub page fixed #105	<u>https://github.com/sugarlabs/www-v2/pull/105</u>
	changed svgs and image #114	<u>https://github.com/sugarlabs/www-v2/pull/114</u>

	added matrix page #115	https://github.com/sugarlabs/www-v2/pull/115
	v3 and v4 added #117	https://github.com/sugarlabs/www-v2/pull/117
	Color Detector For Music Blocks #4569	https://github.com/sugarlabs/musicblocks/pull/4596

Structure of the Codebase

```

www-v2/
├─ .github/           # GitHub-specific configurations (e.g., workflows)
├─ docs/              # Documentation files
├─ public/            # Static assets like images and fonts
├─ src/               # Source code
│  └─ components/     # Reusable UI components
│  └─ layouts/         # Page layout components
│  └─ pages/           # Page components corresponding to routes
│  └─ styles/          # Global and component-specific styles
│  └─ data/            # JSON and Markdown content
├─ .eslintrc.js       # ESLint configuration
├─ .gitignore         # Specifies files and directories to ignore in Git
├─ .markdownlint.json # Markdown linting configuration
├─ .prettierignore    # Prettier ignore file
├─ .prettierrc        # Prettier configuration
├─ CNAME              # Custom domain configuration for GitHub Pages
├─ LICENSE            # Project license (GPL-3.0)
├─ README.md          # Project overview and setup instructions
├─ components.json    # Component metadata
├─ eslint.config.js   # Additional ESLint configuration
├─ index.html         # Entry point HTML file
├─ package-lock.json  # Exact versions of installed dependencies
├─ package.json       # Project metadata and dependencies
├─ tsconfig.app.json  # TypeScript configuration for the app
├─ tsconfig.json      # Base TypeScript configuration
├─ tsconfig.node.json # TypeScript configuration for Node.js
└─ vite.config.ts     # Vite bundler configuration

```

Key Highlights:

- **Tech Stack:** The project utilizes TypeScript, React, Tailwind CSS, and Vite for bundling. [GitHub](#)
- **Source Directory (src/):**
 - components/: Houses reusable UI elements.
 - layouts/: Contains layout components that define the structure of pages.
 - pages/: Includes page components that correspond to different routes.
 - styles/: Stores global and component-specific styles.
 - data/: Contains JSON and Markdown files used for content. [sugarlabs.org](#)
- **Configuration Files:**
 - .eslintrc.js & eslint.config.js: Define linting rules for the project.
 - .prettierrc & .prettierignore: Configure code formatting standards.
 - tsconfig*.json: Manage TypeScript compiler options.
 - vite.config.ts: Sets up the Vite bundler for development and production builds. [wiki.sugarlabs.org+1GitHub+1](#)
- **Public Directory (public/):** Contains static assets that are served directly without processing.
- **Documentation (docs/):** Provides additional information and guides related to the project.

Learnings & Impact

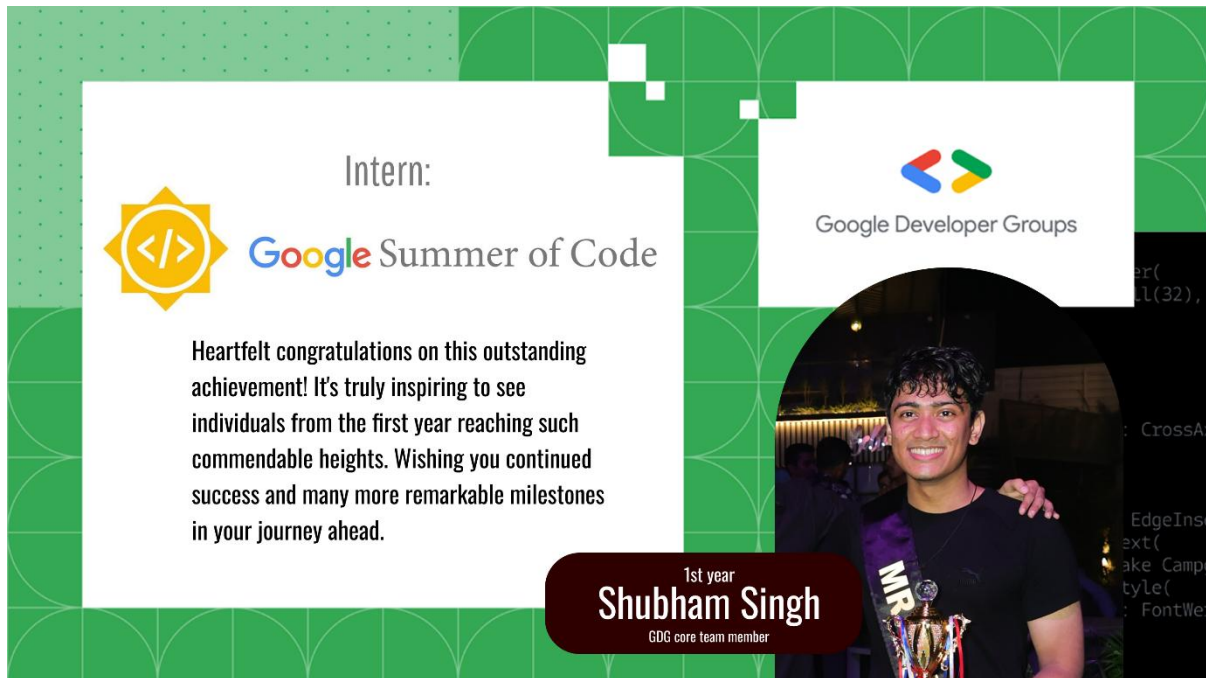
- Learned how to manage and contribute to large-scale open-source repositories.
- Enhanced leadership, communication, and collaboration skills by mentoring contributors.
- Gained deep hands-on experience in frontend architecture and design systems.
- Contributed to a project that directly impacts the global educational community.

Achievements

- **Selected for Google Summer of Code (GSoC) 2025** with Sugar Labs.
 - **Lead designer** credited in the www-v2
 - Contributed to an official nonprofit site with a worldwide audience.
-

Conclusion

This website redevelopment for Sugar Labs was both a technically and socially meaningful project. It allowed me to apply my skills in **web development, UI/UX design, accessibility, and team collaboration** on a global platform. Through this initiative, I have not only improved the face of Sugar Labs online but also deepened my passion for open-source and educational tools.

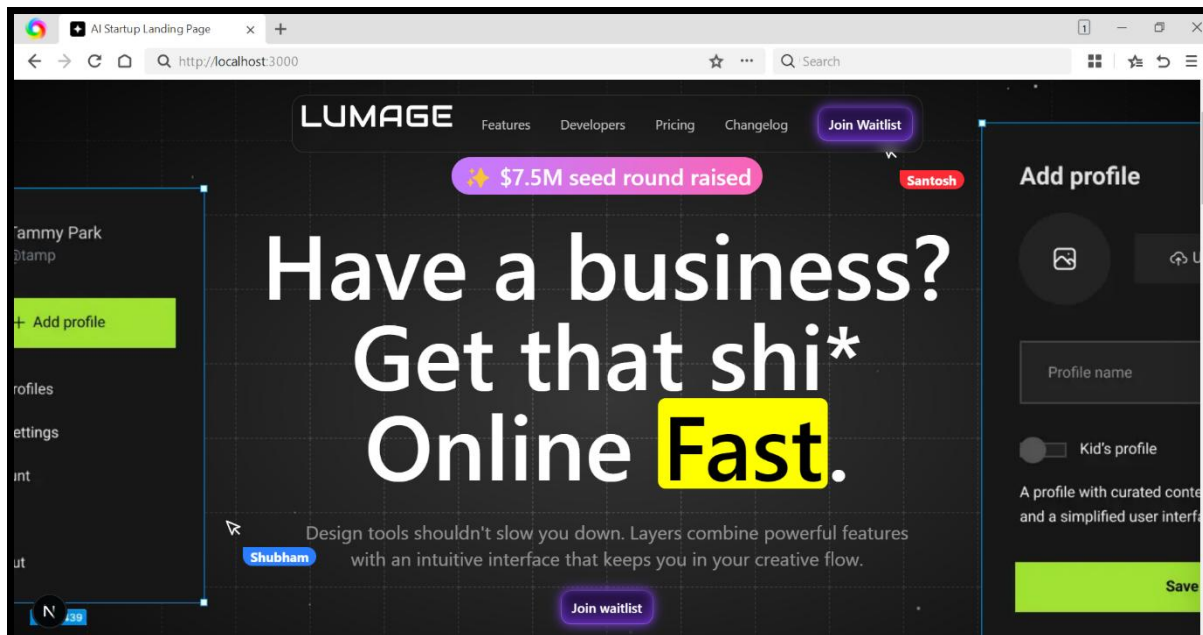


Please go the next page as well :)

Future Interests: (goals to accomplish in second year)

Will start working on:

- 1) BSP Publications Full stack project.
- 2) Lumage Agency (building the startup)



- 3) Cryptic Agency (building the startup) (WEB 3 based business)
- 4) SugarLabs' Webpage addition for ChatBot, SearchFunctionality, PaymentGateway.
- 5) Open Source contribution in Google Deepmind
- 6) A research paper on LLMs, AI, ML.
- 7) Completing the internship of GSoC.